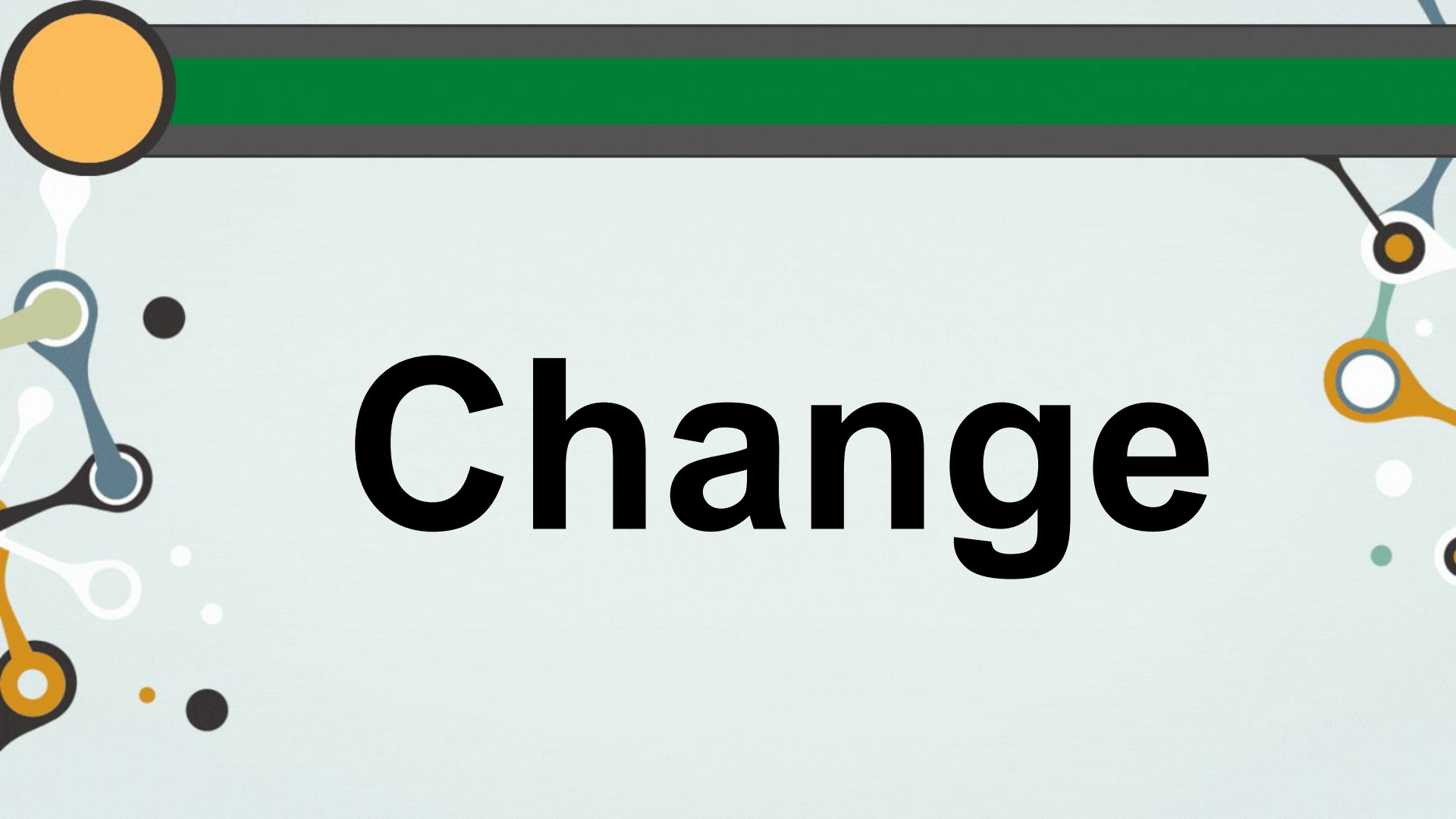
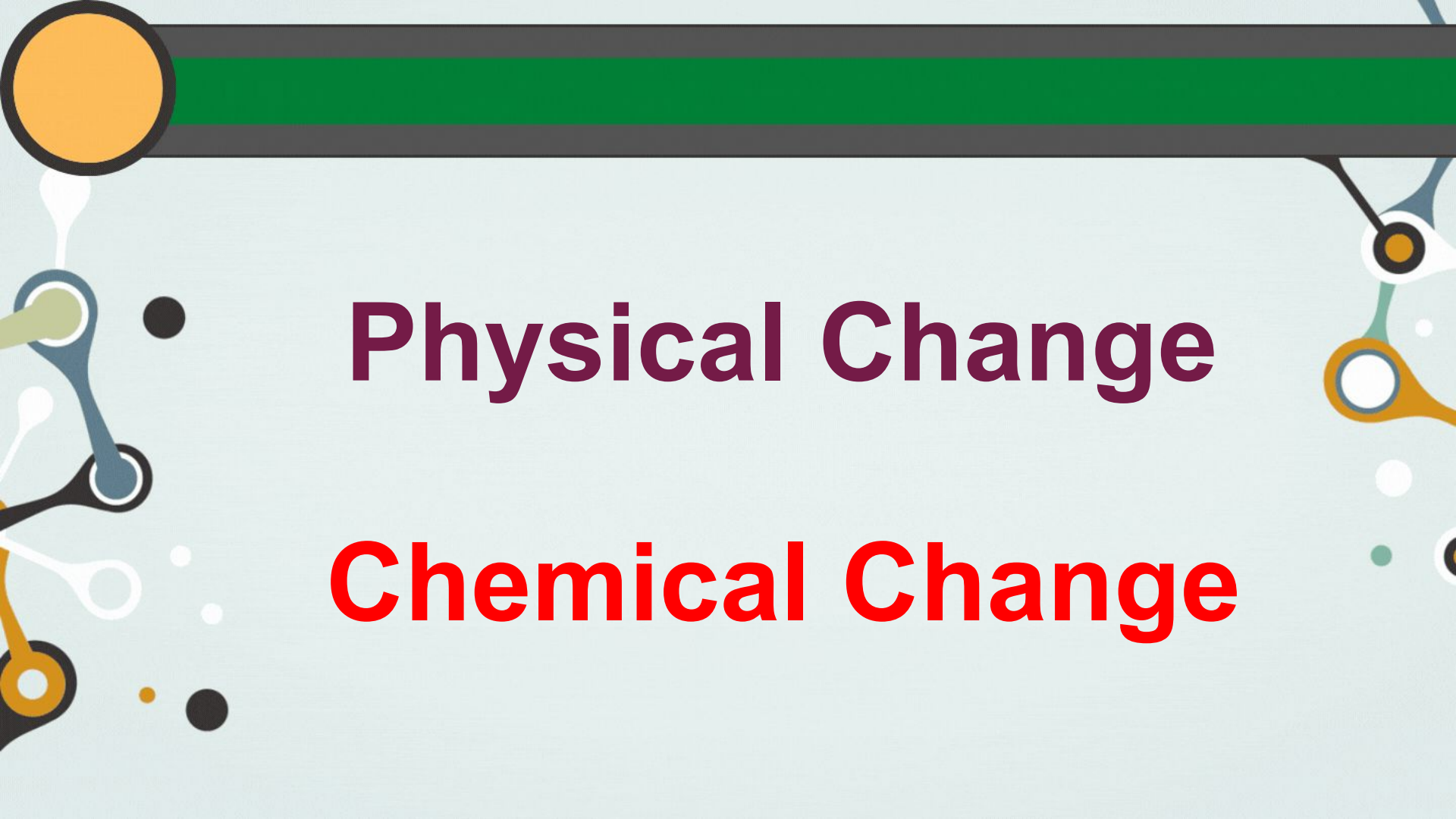


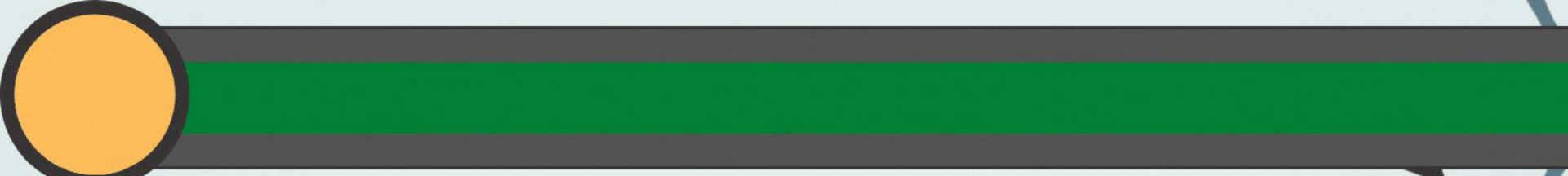
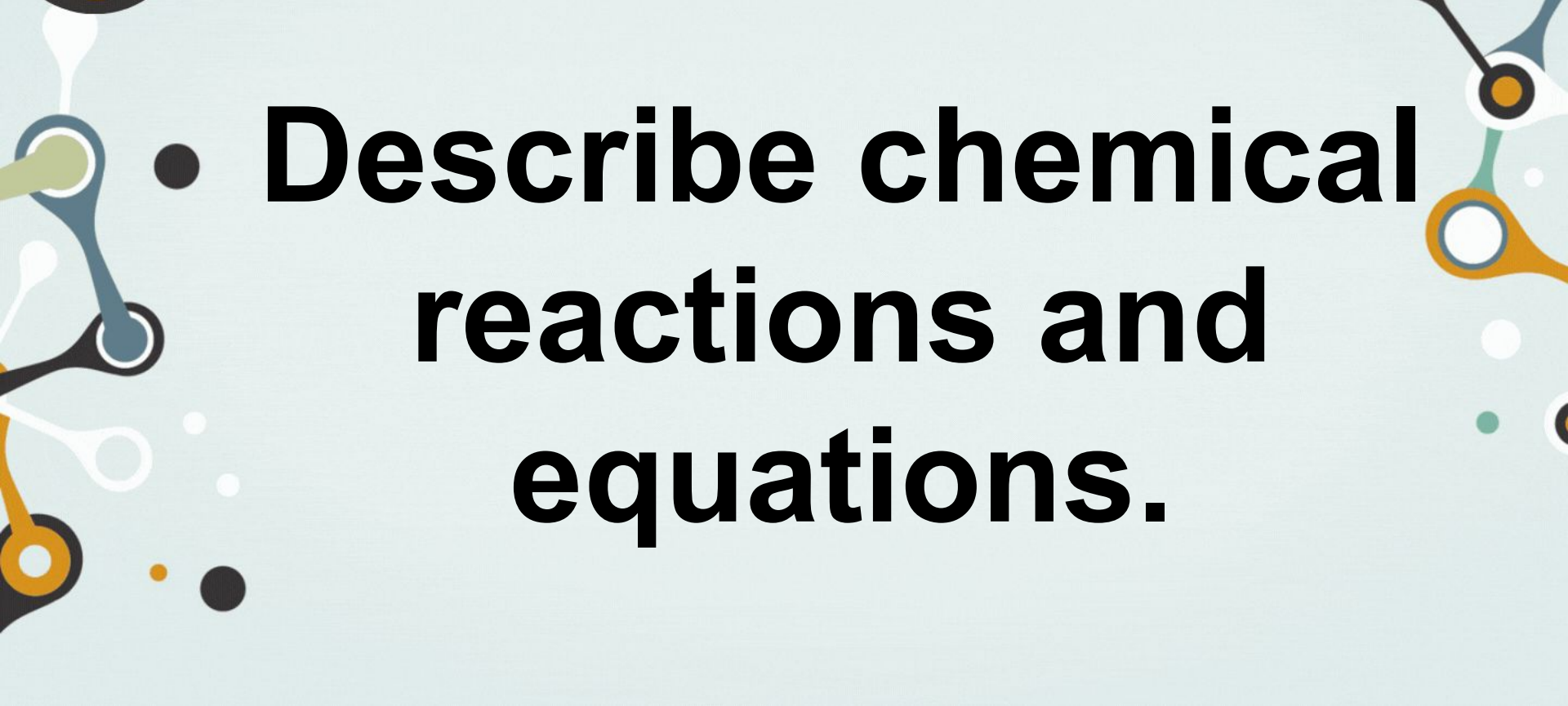
The image features a light blue background with a decorative header at the top consisting of a dark grey bar, a green bar, and a large orange circle on the left. On the left and right sides, there are abstract, stylized molecular or network-like structures in blue, orange, and white. The word "Change" is centered in a large, bold, black font.

Change

The image features a decorative header at the top with a large orange circle on the left, a green horizontal bar, and a dark grey line. On the right side of the header, there is a stylized molecular structure with blue and yellow spheres. The background is light blue with various molecular-like graphics, including blue and yellow spheres connected by lines, and small black and white dots. The text "Physical Change" is written in a large, bold, purple font.

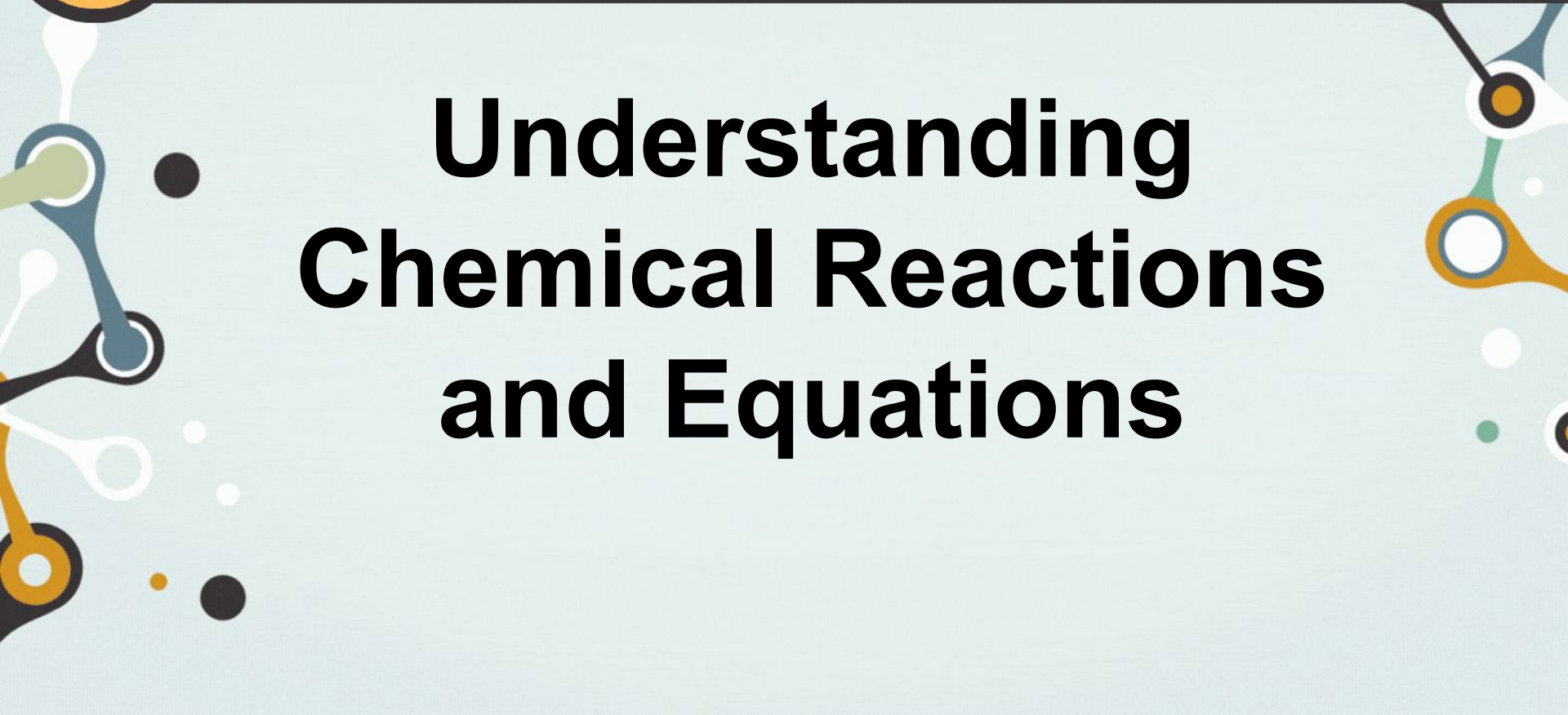
Physical Change

Chemical Change

- 
- 
- **Describe chemical reactions and equations.**



Understanding Chemical Reactions and Equations



Understanding Chemical Reactions and Equations

A **chemical reaction** is a process through which one or more substances are changed into different substances. The original substances are called ***reactants***, and the resulting substances are referred to as ***products***.

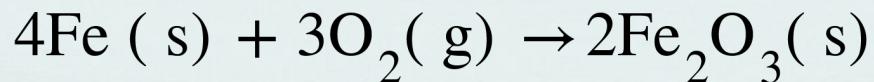


Understanding Chemical Reactions and Equations

Representing Chemical Reactions

Chemical reactions are represented by ***chemical equations***. Consisting of symbols and formulas, chemical equations show what happens during a reaction.

The equations present the the identities and relative amounts of the reactants and products involved in a chemical reaction.



Understanding Chemical Reactions and Equations

Symbols in Chemical Equations

Symbol	Function
$+$	Written between the symbols and/or formulas of reactants or products Left side: read as “combines with” or “reacts with” Right side: read as “and”
$\rightarrow$	Separates reactants from products Read as “yields” or “produces”

Understanding Chemical Reactions and Equations

Symbols in Chemical Equations

Symbol	Function
(s)	Indicates a solid reactant or product
(l)	Indicates a liquid reactant or product

Understanding Chemical Reactions and Equations

Symbols in Chemical Equations

Symbol	Function
<i>(g)</i>	Indicates a gaseous reactant or product
<i>(aq)</i>	Indicates that the reactant or product is in an aqueous solution (homogeneously dispersed in water)

Understanding Chemical Reactions and Equations

Symbols in Chemical Equations

Symbol	Function
Δ →	Indicates that heat must be supplied to reactants before a reaction occurs
catalyst →	The substance written above the arrow is a catalyst or a solvent

Understanding Chemical Reactions and Equations

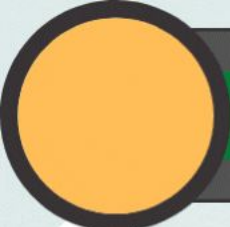


The big numbers are called ***coefficients***, which give an idea of the number of molecules or formula units of a substance that participate in the reaction.

Understanding Chemical Reactions and Equations

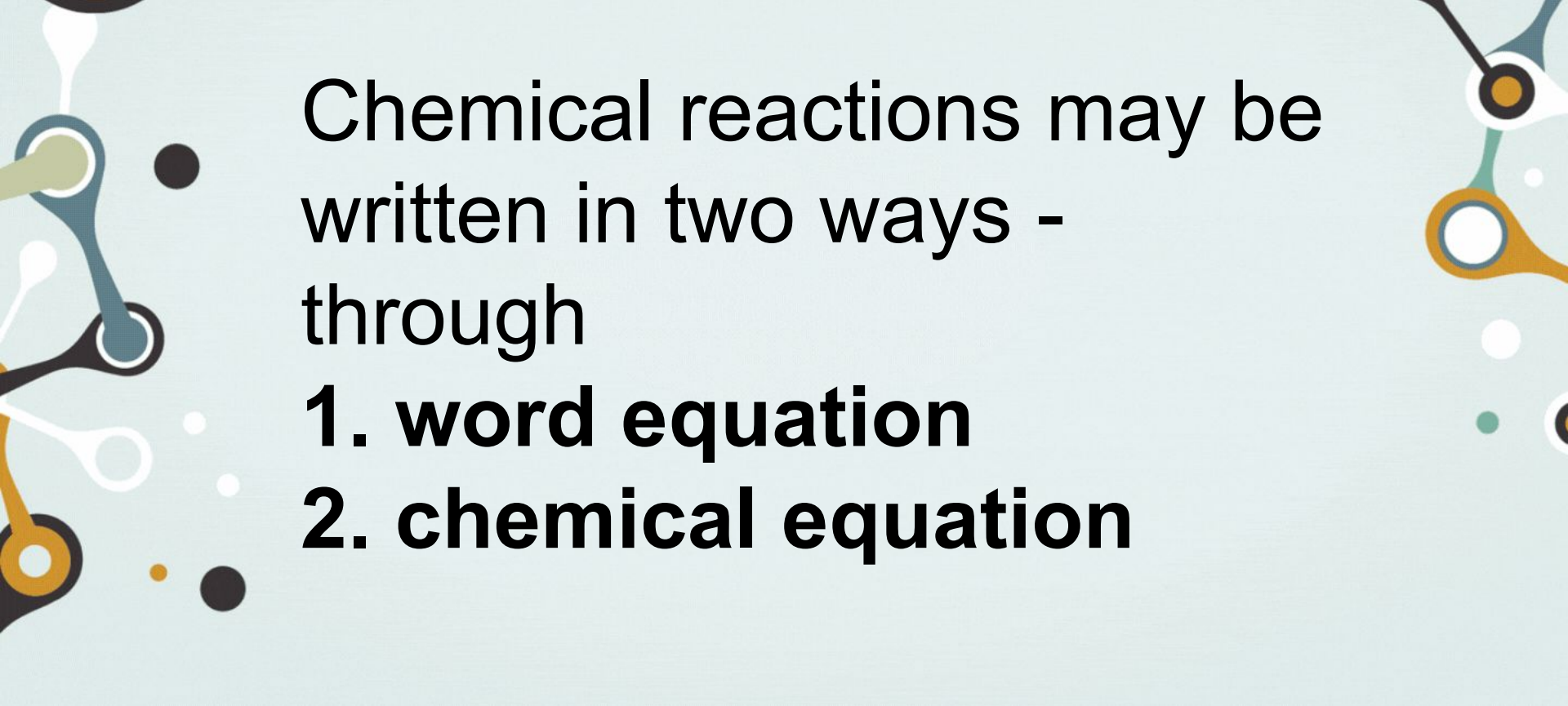


The small numbers are termed ***subscripts***, which tell how many atoms are present in a specific compound.



Understanding Chemical Reactions and Equations

Chemical reactions may be written in two ways - through

- 1. word equation**
 - 2. chemical equation**
- 

Understanding Chemical Reactions and Equations

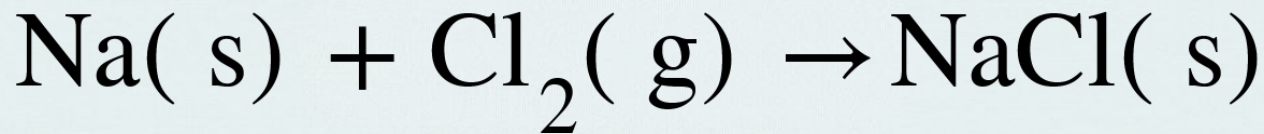
Word Equations

- A **word equation** presents the reactants and products of a chemical reaction using words. It has only qualitative meaning. It only identifies the reactants and products involved in a reaction, and not their quantities.

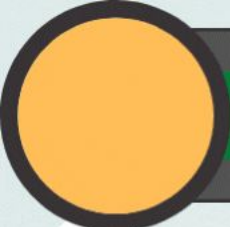
- hydrogen gas + chlorine gas $\rightarrow$ hydrogen chloride gas

Understanding Chemical Reactions and Equations

Translate the chemical equation into word equation.



solid sodium metal + chlorine gas $\rightarrow$ solid sodium chloride



Understanding Chemical Reactions and Equations

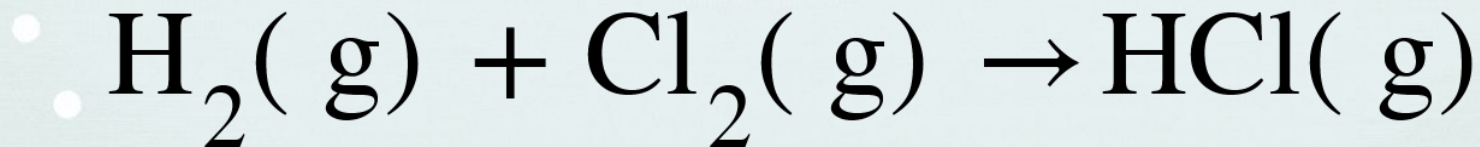
Chemical Equations

A word equation can be converted to a chemical equation by replacing the names of the reactants and products with their corresponding symbols or formulas. Solids, liquids, and gases are usually labeled as **(s)**, **(l)**, and **(g)**, respectively. Solids include the ***precipitates***, which are insoluble substances that may form in a reaction. Substances dissolved in water are considered ***aqueous*** and are labeled **(aq)**.

Understanding Chemical Reactions and Equations

Translate the word equation into chemical equation.

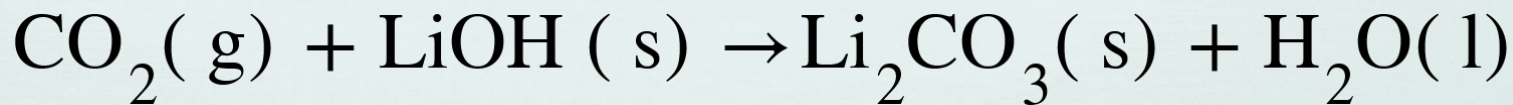
hydrogen gas + chlorine gas $\rightarrow$ hydrogen chloride gas



Understanding Chemical Reactions and Equations

Translate the word equation into chemical equation.

Carbon dioxide gas reacts with solid lithium hydroxide (LiOH) to produce solid lithium carbonate (Li₂CO₃) and water.

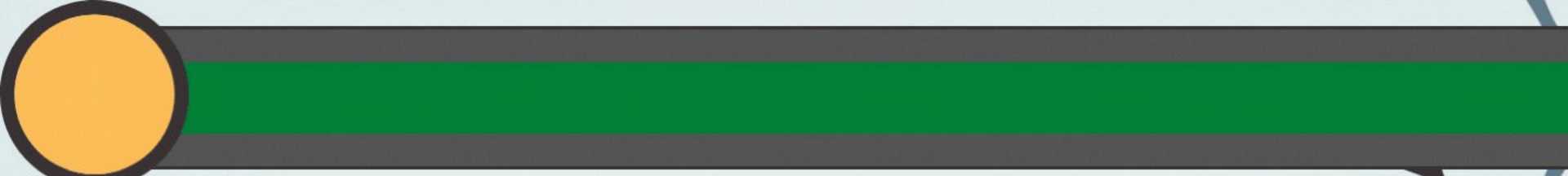


Understanding Chemical Reactions and Equations

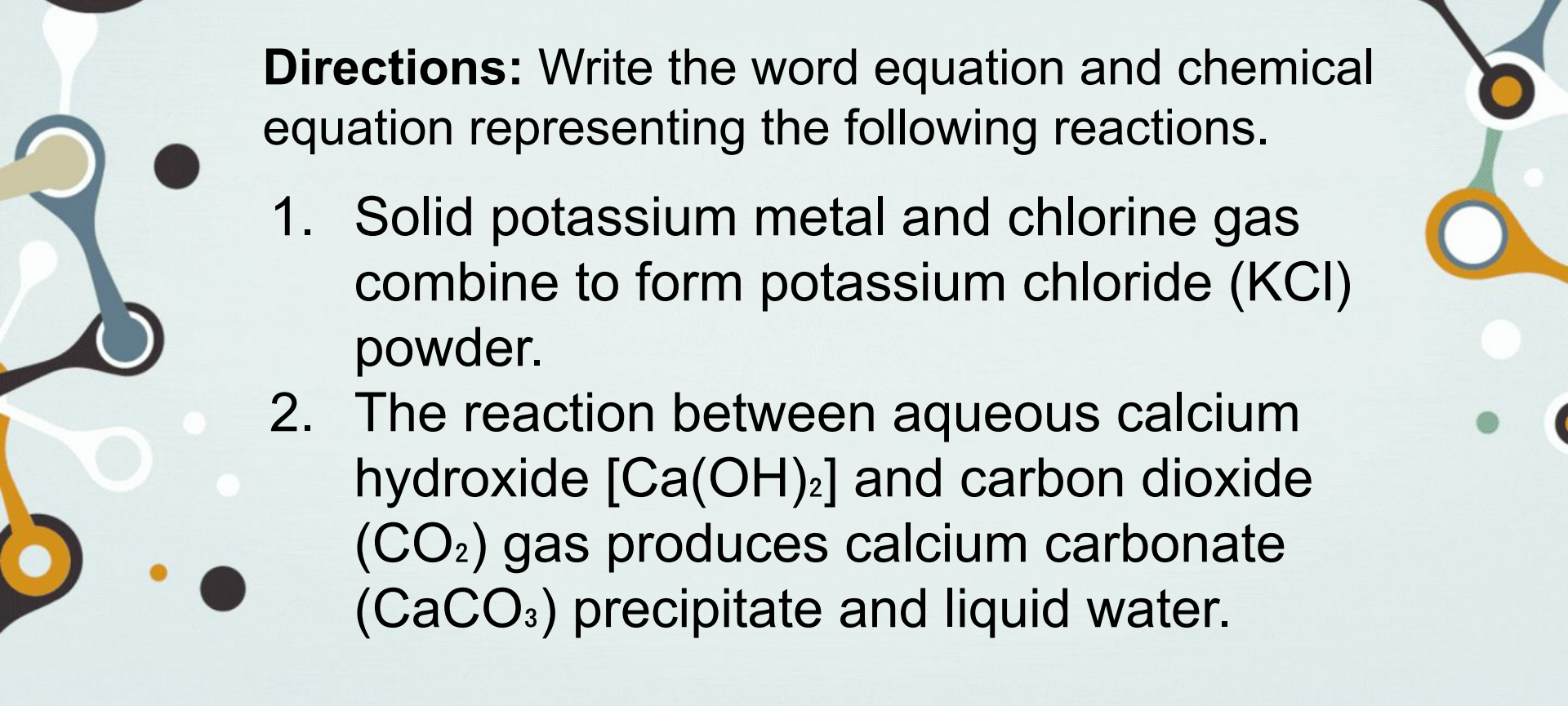
Translate the word equation into chemical equation.

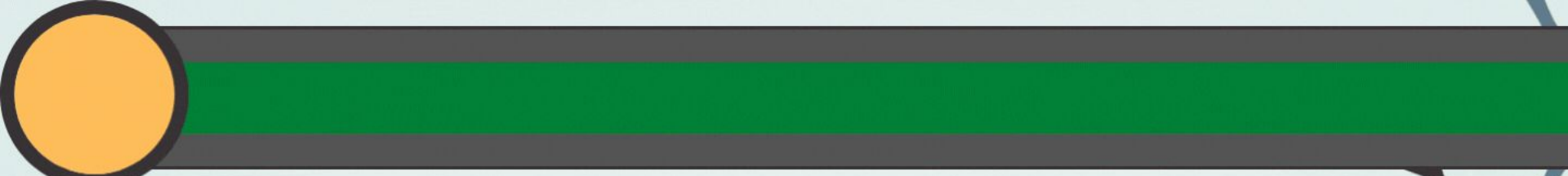
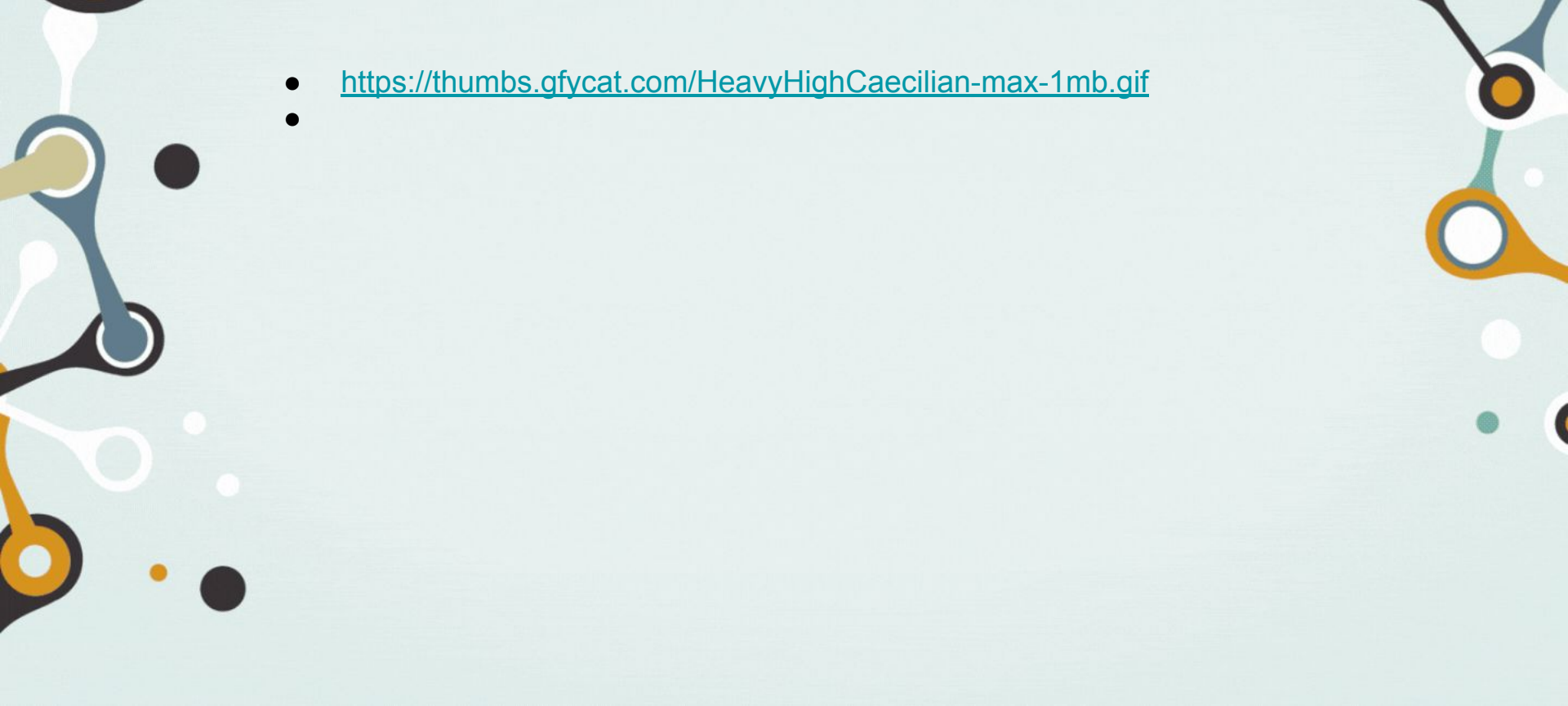
Solid potassium chlorate (KClO_3) decomposes to produce solid potassium chloride and oxygen gas.





Directions: Write the word equation and chemical equation representing the following reactions.

1. Solid potassium metal and chlorine gas combine to form potassium chloride (KCl) powder.
 2. The reaction between aqueous calcium hydroxide $[\text{Ca}(\text{OH})_2]$ and carbon dioxide (CO_2) gas produces calcium carbonate (CaCO_3) precipitate and liquid water.
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